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Section :-

(0)

Submitted To :-

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Assignment No. 4 :-

Exercise 10.5.1 :-

Positive integers :-

$$n \in \mathbb{Z}^+ = \{1, 2, 3, \dots\}$$

Inductive hypothesis :-

put n in (n)

Non-negative integers :-

$$n \in \mathbb{Z} = \{0, 1, 2, 3, 4, \dots\}$$

Question 10.3 :-

Let $P(n)$ be the statement that $1^2 + 2^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$ for the positive integer n .

(a)

What is the statement $P(1)$?

Put $n=1$ in $P(1)$

$$1^2 = \frac{1(1+1)(2 \cdot 1 + 1)}{6}$$

$$1^2 = \frac{1(1+1)(2(1)+1)}{6}$$

$$1^2 = \frac{1(2)(2+1)}{6}$$

$$1^2 = \frac{2(3)}{6}$$

$$1^2 = \frac{6}{6} = 1$$

(b)

Show that $P(1)$ is true, completing the basic step of the proof.

$$\text{L.H.S of } P(1) = 1^2 = 1$$

$$\text{R.H.S of } P(1) = \frac{(1)(2)(3)}{6} = \frac{6}{6} = 1$$

$P(1)$ is true

(C). What is the inductive hypothesis of a proof that $P(n)$ is true for all positive integers n ? → (K) put

$P(K)$ is true

i.e :

$$1^2 + 2^2 + 3^2 + \dots + K^2 = \frac{K(K+1)(2K+1)}{6}$$

(d) What do you need to prove in the inductive step?

Assume inductive hypothesis is true.

$$1^2 + 2^2 + \dots + K^2 = \frac{K(K+1)(2K+1)}{6}$$

$$1^2 + 2^2 + \dots + K^2 + (K+1)^2 = \frac{(K+1)(K+1+1)(2K+3)}{6}$$

$$1^2 + 2^2 + \dots + K^2 + (K+1)^2 = \frac{(K+1)(K+2)(2K+3)}{6}$$

(e)

Complete the inductive step, identifying where you use the inductive hypothesis.

$$\frac{k(k+1)(2k+1)}{6} + (k+1)^2$$

$$= \frac{k(k+1)(2k+1) + 6(k+1)^2}{6}$$

$$= \frac{k+1[k(2k+1) + 6(k+1)]}{6}$$

$$= \frac{(k+1)[2k^2 + k + 6k + 6]}{6} \quad (f)$$

$$= \frac{(k+1)(2k^2 + 7k + 6)}{6}$$

$$= \frac{(k+1)(2k^2 + 7k + 6)}{6}$$

$$= \frac{(k+1)(2k^2 + 4k + 3k + 6)}{6}$$

$$= \frac{(k+1)(2k(k+1) + 3(k+2))}{6}$$

$$= \frac{(k+1)(k+2)(2k+3)}{6}$$

Explain why these steps show that this formula is true whenever n is a positive integer.

Both the basis and Inductive steps are true (Part (a) and b). So, by the mathematical Induction, the given statement is true.

Question No 4:-

Let $P(n)$ be the statement that $1^3 + 2^3 + \dots + n^3 = \frac{(n(n+1))^2}{4}$ for the positive integer n .

(a)

What is the statement $P(1)$?

$$1^3 + 2^3 + \dots + n^3 = \frac{(n(n+1))^2}{4}$$

$$P(1) = 1^3 = \frac{(1(1+1))^2}{4}$$

$$= 1^3 = \left(\frac{(1)(2)}{2} \right)^2$$

$$= 1^3 = \left(\frac{2}{2} \right)^2$$

$$= 1^3 = 1^2$$

$$\boxed{1 = 1} \text{ proved}$$

(b)

Show that $P(1)$ is true, completing the basic step of the proof.

$$1^3 + 2^3 + \dots + n^3 = \left(\frac{n(n+1)}{2} \right)^2$$

$$1^3 = \left(\frac{1(1+1)}{2} \right)^2$$

$$1^3 = \left(\frac{1(2)}{2} \right)^2$$

$$1^3 = \left(\frac{2}{2} \right)^2$$

$$1^3 = 1^2$$

$$\boxed{1 = 1} \text{ proved}$$

(c)

What is the Inductive hypothesis?

$$1^3 + 2^3 + \dots + k^3 = \left(\frac{k(k+1)}{2} \right)^2$$

(d)

What do you need to prove in the inductive step?

$$1^3 + 2^3 + \dots + k^3 = \left(\frac{k(k+1)}{2} \right)^2$$

$$1^3 + 2^3 + \dots + k^3 + (k+1)^3 = \left(\frac{(k+1)((k+1)+1)}{2} \right)^2$$

$$k^3 + (k+1)^3 = \left(\frac{(k+1)(k+2)}{2} \right)^2$$

(c)

Complete the inductive step, identifying where you use the inductive hypothesis.

$$1^3 + 2^3 + 3^3 + \dots + k^3 + (k+1)^3 = \left(\frac{(k+1)(k+2)}{2} \right)^2$$

$$\left(\frac{k(k+1)}{2} \right)^2 + (k+1)^3$$

$$\left(\frac{k(k+1)}{2} \right)^2 + (k+1)^3$$

$$\frac{k^2(k+1)^2}{4} + (k+1)^3$$

$$\frac{k^2(k+1)^2 + 4(k+1)^3}{4}$$

$$\frac{(k+1)(k^2 + 4k + 4)}{4}$$

$$\frac{(k+1)^2(k^2 + 2k(2) + (2)^2)}{4}$$

$$\frac{(k+1)^2(k+2)^2}{4}$$

$$\boxed{\left(\frac{(k+1)(k+2)}{2} \right)^2}$$

Ans

$$\therefore (k+2)^2$$

$$k^2 + 4 + 2k(2)$$

$$k^2 + 4k + 4$$

(f)

Explain why these steps show that this formula is true whenever n is a positive integer.

Both the base and Inductive steps are true. Part b and e. So, by the mathematical induction the given statement is true.

Question 10.5 :-

Prove that $1^2 + 3^2 + 5^2 + \dots + (2n+1)^2 = (n+1)(2n+1)(2n+3)/3$ whenever n is a nonnegative integer.

Sol :-

$$n \in \{0, 1, 2, 3, 4, \dots\}$$

Basic step :- put $n=0$.

$$1^2 = \frac{(n+1)(2n+1)(2n+3)}{3}$$

$$1^2 = \frac{(0+1)(2(0)+1)(2(0)+3)}{3}$$

$$1^2 = \frac{(1)(1)(3)}{3}$$

$$1^2 = \frac{(1)(1)(3)}{3}$$

$$1^2 = (1)$$

$$1 = 1$$

True.

Inductive Step :

Inductive hypothesis:

$$1^2 + 3^2 + 5^2 + \dots + (2k+1)^2 = \frac{(k+1)(2k+1)(2k+3)}{3}$$

K next step :

$$1^2 + 3^2 + 5^2 + \dots + (2k+1)^2 + (2k+3)^2 = \frac{(k+2)(2k+3)(2k+5)}{3}$$

$$\left[1^2 + 3^2 + 5^2 + \dots + (2k+1)^2 \right] + (2k+3)^2$$

$$\frac{(k+1)(2k+1)(2k+3)}{3} + \cancel{\frac{(2k+3)^2}{1}}$$

$$\frac{(k+1)(2k+1)(2k+3) + 3(2k+3)^2}{3}$$

$$(2k+3) \left[\frac{(k+1)(2k+1) + 3(2k+3)}{3} \right]$$

$$= (2k+3) \left[\frac{2k^2 + k + 2k + 1 + 6k + 9}{3} \right]$$

$$= (2k+3) \left(\frac{2k^2 + 3k + 10}{3} \right)$$

$$= (2k+3) \left[\frac{2k^2 + 5k + 4k + 10}{3} \right]$$

$$= (2k+3) \left[\frac{2k^2(2k+5) + 10(2k+5)}{3} \right]$$

$$= (2k+3) \left[\frac{(k+2)(2k+5)}{3} \right]$$

$$= \boxed{\frac{(k+2)(2k+3)(2k+5)}{3}} \quad \text{True}$$

So, Inductive Step is true $\forall k \geq 0$

By mathematical Induction:

$$1^2 + 3^2 + 5^2 + \dots + (2n+1)^2 = \frac{(n+1)(2n+1)(2n+5)}{3}$$

is true for every n is a non-negative integers.

Question :- Q 6 :-

Prove that $1 \cdot 1! + 2 \cdot 2! + \dots + n \cdot n! = (n+1)! - 1$
whenever n is a positive integer.

Sol :-

$$n \in \{1, 2, 3, \dots\}$$

Basic Step :- put $n = 1$

$$\begin{aligned} 1 \cdot 1! &= (n+1)! - 1 \\ &= (1+1)! - 1 \\ &= 2! - 1 \\ &= 2 - 1 \end{aligned}$$

$$1 = 1 \quad \text{True.}$$

Inductive Step :-

Inductive hypothesis :-

$$1 \cdot 1! + 2 \cdot 2! + \dots + k \cdot k! = (k+1)! - 1$$

k next step :-

$$1 \cdot 1! + 2 \cdot 2! + \dots + (k+1)(k+1)! = (k+2)! - 1$$

L.H.S :-

$$\begin{aligned} 1 \cdot 1! + 2 \cdot 2! + \dots + (k+1)(k+1)! \\ (k+1)! - 1 + (k+1)(k+1)! \end{aligned}$$

$$\begin{aligned}
 &= (k+1)! + (k+1)(k+1)! - 1 \\
 &= (k+1)! [1 + k + 1] - 1 \\
 &= (k+1)! (k+2) - 1 \\
 &= (k+2)(k+1)! - 1 \\
 &= \boxed{(k+2)! - 1} \quad \text{True} \quad 7! = 7 \times 6! \quad \boxed{?}
 \end{aligned}$$

Hence proved.

Question 10.7.3 -

Prove that $3 + 3 \cdot 5 + 3 \cdot 5^2 + \dots + 3 \cdot 5^n = \frac{3(5^{n+1} - 1)}{4}$ whenever n is a non-negative integer.

Base class :

$$3 + 3 \cdot 5 + 3 \cdot 5^2 + \dots + 3 \cdot 5^n + \frac{3(5^{n+1} - 1)}{4}$$

Put $n=0$

$$P(0) = 3 = \frac{3(5^{0+1} - 1)}{4}$$

$$= 3 = \frac{3(5^{0+1} - 1)}{4}$$

$$= 3 = \frac{3(5^1 - 1)}{4}$$

$$= 3 = \frac{3 \cdot 4}{4}$$

$$3 = 3 \quad \text{True.}$$

Inductive Step :

put $n = k$

$$3 + 3 \cdot 5 + 3 \cdot 5^2 + \dots + 3 \cdot 5^k = \frac{3(5^{k+1} - 1)}{4}$$

$$3 + 3 \cdot 5 + 3 \cdot 5^2 + \dots + 3 \cdot 5^k = 3 \cdot \frac{(5^{k+1} - 1)}{4}$$

$$3 + 3 \cdot 5 + 3 \cdot 5^2 + \dots + 3 \cdot 5^k + 3 \cdot 5^{k+1} = 3 \cdot \frac{(5^{k+2} - 1)}{4}$$

$$3 + 3 \cdot 5 + 3 \cdot 5^2 + \dots + 3 \cdot 5^k + 3 \cdot 5^{k+1} = 3 \cdot \frac{(5^{k+2} - 1)}{4}$$

$$= 3 \cdot \frac{(5^{k+1} - 1)}{4} + 3 \cdot 5^{k+1}$$

$$= 3 \cdot \frac{(5^{k+1} - 1)}{4} + 4 \cdot 3 \cdot 5^{k+1}$$

$$= 3 \cdot \frac{(5^{k+1} - 1 + 4 \cdot 5^{k+1})}{4}$$

$$= 3 \cdot \frac{(5^{k+1} + 4 \cdot 5^{k+1} - 1)}{4}$$

$$= 3 \cdot \frac{(5^{k+1}(1+4) - 1)}{4}$$

$$= 3 \cdot \frac{(5^{k+1}(5) - 1)}{4}$$

$$= \frac{3(5^{k+2} - 1)}{4} \quad \text{True}$$

Hence proved.

Question 20 & :-

Prove that $2 + 2 \cdot 7 + 2 \cdot 7^2 + \dots + 2(-7)^n = \frac{(1 - (-7)^{n+1})}{4}$
whenever n is a non-positive integers.

Base Case : put $n=0$

$$p(0) = 2 = \frac{(1 - (-7)^{0+1})}{4}$$

$$= 2 = \frac{(1 - (-7)^{0+1})}{4}$$

$$= 2 = \frac{(1 - (-7)^1)}{4}$$

$$= 2 = \frac{(1 + 7)}{4}$$

$$2 = \frac{8}{4}$$

$$2 = 2 \quad \text{True}$$

Inductive Step:

put $n = k$

$$2 - 2 \cdot 7 + 2 \cdot 7^2 + \dots + 2(-7)^n = \frac{(1 - (-7)^{n+1})}{4}$$

$$2 - 2 \cdot 7 + 2 \cdot 7^2 + \dots + 2(-7)^k = \frac{(1 - (-7)^{k+1})}{4}$$

$$2 - 2 \cdot 7 + 2 \cdot 7^2 + \dots + 2(-7)^k + 2(-7)^{k+1} = \frac{(1 - (-7)^{k+2})}{4}$$

$$2 - 2 \cdot 7 + 2 \cdot 7^2 + \dots + 2(-7)^k + 2(-7)^{k+1} = \frac{(1 - (-7)^{k+1})}{4}$$

$$= \frac{(1 - (-7)^{k+1})}{4} + 4 \cdot 2(-7)^{k+1}$$

$$= \frac{(1 - (-7)^{k+1}) + 8(-7)^{k+1}}{4}$$

$$= \frac{(1 - (-7)^{k+1}) + 8(-7)^{k+1}}{4}$$

$$= \frac{(1 - [(-7)^{k+1} - 8(-7)^{k+1}])}{4}$$

$$= \frac{(1 - [(-7)^{k+1} \cdot -7])}{4}$$

$$= \frac{(1 - [-7^{k+1} - 1])}{4}$$

$$= \boxed{\frac{1 - (-7^{k+2})}{4}} \quad \text{True}$$

Hence proved.

Question 20 Q. 9.

(Q. 9)

Find a formula for the sum of first n even positive integers.

$$2, 4, 6, \dots, 2n$$

$$2 + 4 + 6 + \dots + 2n$$

By using arithmetic Series:

$$S_n = \frac{n}{2} (2a_1 + (n-1)d)$$

$$\text{Here } a_1 = \boxed{2}$$

$$d = 4 - 2 = \boxed{2}$$

$$S_n = \frac{n}{2} (2(2) + (n-1)2)$$

$$= \frac{n}{2} (4 + (n-1)2)$$

$$= \frac{n}{2} (4 + 2n - 2)$$

$$= \frac{n}{2} (2 + 2n)$$

$$= \frac{n}{2} (2 + 2n)$$

$$= \frac{n}{2} (2(1+n))$$

$$= n(1+n)$$

$$= \boxed{n(n+1)}$$

(b.)
Prove that formula that you conjectured
in part (a)

Base step : $p(n) = 1$

$$p(n) = 2 = n(n+1)$$

$$p(1) = 2 = 1(1+1)$$

$$2 = 2 \text{ True.}$$

Inductive step :

put $n = k+1$

$$2 + 4 + 6 + \dots + 2n = n(n+1)$$

$$2 + 4 + 6 + \dots + 2k = k(k+1)$$

$$2 + 4 + 6 + \dots + 2k + 2(k+1) = (k+1)(k+1+1)$$

$$= (k+1)(k+2)$$

$$2 + 4 + 6 + \dots + 2k + 2(k+1) = (k+1)(k+2)$$

$$k(k+1) + 2(k+1)$$

$$= (k+1)(k+2)$$

Question No 10 :-

(a)

Find the formula for $\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{n(n+1)}$

$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{n(n+1)}$$

As :-

$n = 1$ and $n = 2 \Rightarrow \text{Sum}$

$$\frac{1}{2} + \frac{1}{6} = \frac{6+2}{12} = \frac{8}{12} = \frac{2}{3}$$

and $n = 3 \Rightarrow \text{Sum}$

$$\frac{1}{2} + \frac{1}{6} + \frac{1}{12} = \frac{6+2+1}{12} = \frac{9}{12} = \frac{3}{4}$$

$n = 4 \Rightarrow \text{Sum}$

$$\frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \frac{1}{20} = \frac{30 + (10) + 5 + 3}{60} = \frac{48}{60} = \frac{4}{5}$$

So;

$$\frac{1}{2 \cdot 3} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{n(n+1)} = \frac{n}{n+1}$$

(b)

Prove that the formula that you conjectured in part (a)

Base Case :

$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{n(n+1)} = \frac{n}{n+1}$$

Put $n = 1$

$$P(1) = \frac{1}{1 \cdot 2} = \frac{1}{1+1}$$

$$= \frac{1}{2} = \frac{1}{2} \quad \text{True.}$$

Inductive Step :

put $n = k$

$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{n(n+1)} = \frac{n}{n+1}$$

$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{k(k+1)} = \boxed{\frac{k}{k+1}}$$

$$\frac{1}{1 \cdot 2} + \frac{1}{1 \cdot 3} + \dots + \frac{1}{k(k+1)} + \frac{1}{(k+1)(k+2)} = \frac{k+1}{k+2}$$

$$= \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{k(k+1)} + \frac{1}{(k+1)(k+2)} = \frac{k}{k+1}$$

$$= \frac{k}{k+1} + \frac{1}{(k+1)(k+2)}$$

$$= \frac{k(k+2) + 1}{(k+1)(k+2)}$$

$$= \frac{k^2 + 2k + 1}{(k+1)(k+2)}$$

$$= \frac{(k+1)^2}{(k+1)(k+2)}$$

$$\boxed{\frac{k+1}{k+2}} \text{ True}$$

Hence proved.

Question 10 & 11 :-

(a)

Find a formula for $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots + \frac{1}{2^n}$

$$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots + \frac{1}{2^n}$$

By Geometric Progression

$$\text{Sol}^n \quad \frac{a(1-r^n)}{1-r}$$

$$= \frac{1}{2} (1-r^n)$$

$$1-r$$

$$= \frac{1}{2} \left(1 - \frac{1}{2^n}\right)$$

$$\frac{1 - \frac{1}{2^n}}{2}$$

$$\frac{\frac{1}{2} (2^n - 1)}{\frac{2^n}{2^n}}$$

$$\frac{2^n - 1}{2}$$

$$\frac{1}{2} \left(\frac{2^n - 1}{2^n} \right)$$

$$\frac{1}{2}$$

$$= \boxed{\frac{2^n - 1}{2^n}}$$

(b)

Prove that the formula that you conjecture in Part (a).

Base Case :

$$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots + \frac{1}{2^n} = \frac{2^n - 1}{2^n}$$

$$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots + \frac{1}{2^n} = \frac{2^n - 1}{2^n}$$

$$P(1) = \frac{1}{2} = \frac{2^1 - 1}{2^1}$$

$$= \frac{1}{2} = \frac{2^1 - 1}{2^1}$$

$$= \frac{1}{2} = \frac{1}{2} \quad \text{True}$$

Inductive Step :

$$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots + \frac{1}{2^n} = \frac{2^n - 1}{2^n}$$

put $n = k$

$$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots + \frac{1}{2^k} = \frac{2^k - 1}{2^k}$$

$$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{2^k} + \frac{1}{2^{k+1}} = \frac{2^{k+1} - 1}{2^{k+1}}$$

Now :-

$$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots + \frac{1}{2^k} + \frac{1}{2^{k+1}} = \frac{2^k - 1}{2^{k+1}}$$

$$= \frac{2^k - 1}{2^k} + \frac{1}{2^{k+1}}$$

$$= \frac{2(2^k - 1)}{2 \cdot 2^k} + \frac{1}{2^{k+1}}$$

$$= \frac{2(2^k - 1)}{2^{k+1}} + \frac{1}{2^{k+1}}$$

$$= \frac{2(2^k - 1) + 1}{2^{k+1}}$$

$$= \boxed{\frac{2^{k+1} - 1}{2^{k+1}}} \quad \text{True}$$

Hence proved.

Question No 19 :-

Prove that

$$\sum_{j=0}^n \left(-\frac{1}{2}\right)^j = \frac{2^{n+1} + (-1)^n}{3 \cdot 2^n}$$

Sol :-

$$\sum_{j=0}^n \left(-\frac{1}{2}\right)^j = \frac{2^{n+1} + (-1)^n}{3 \cdot 2^n}$$

$$\left(-\frac{1}{2}\right)^0 + \left(-\frac{1}{2}\right)^1 + \left(-\frac{1}{2}\right)^2 + \dots + \left(-\frac{1}{2}\right)^n = \frac{2^{n+1} + (-1)^n}{3 \cdot 2^n}$$

$$P(1) = \frac{(1) - 1}{2} = \frac{2^{n+1} + (-1)^n}{3 \cdot 2^n}$$

$$= \frac{2-1}{2} = \frac{2^{1+1} + (-1)^1}{3 \cdot 2^1}$$

$$= \frac{1}{2} = \frac{2^2 + (-1)}{3 \cdot 2}$$

$$\frac{1}{2} = \frac{4-1}{6}$$

$$= \frac{1}{2} = \frac{1}{2}$$

$$= \frac{1}{2} = \frac{1}{2} \quad \text{True.}$$

Inductive Step :-

put $n = k$

$$\left(-\frac{1}{2}\right)^0 + \left(-\frac{1}{2}\right)^1 + \left(-\frac{1}{2}\right)^2 + \dots + \left(-\frac{1}{2}\right)^n = \frac{2^{n+1} + (-1)^n}{3 \cdot 2^n}$$

$$\left(-\frac{1}{2}\right)^0 + \left(-\frac{1}{2}\right)^1 + \left(-\frac{1}{2}\right)^2 + \dots + \left(-\frac{1}{2}\right)^k = \frac{2^{k+1} + (-1)^k}{3 \cdot 2^k}$$

$$\left(-\frac{1}{2}\right)^0 + \left(-\frac{1}{2}\right)^1 + \left(-\frac{1}{2}\right)^2 + \dots + \left(-\frac{1}{2}\right)^k + \left(-\frac{1}{2}\right)^{k+1} = \frac{2^{k+2} + (-1)^{k+1}}{3 \cdot 2^{k+1}}$$

$$= \frac{2^{k+1} + (-1)^k}{3 \cdot 2^k} + \left(-\frac{1}{2}\right)^{k+1}$$

$$= \frac{2^{k+1} + (-1)^k}{3 \cdot 2^k} + \frac{(-1)^{k+1}}{2^{k+1}}$$

$$= \frac{2^{k+1} + (-1)^k}{3 \cdot 2^k} + \frac{(-1)^{k+1}}{2^k \cdot 2}$$

$$= \frac{1}{2^k} \left(\frac{2^{k+1} + (-1)^k}{3} + \frac{(-1)^{k+1}}{2} \right)$$

$$= \frac{1}{2^k} \left(\frac{2(2^{k+1} + (-1)^k) + 3(-1)^{k+1}}{3 \cdot 2} \right)$$

$$= \frac{1}{2^k} \left(\frac{2^{k+2} + 2(-1)^k + 3(-1)^{k+1}}{3 \cdot 2} \right)$$

$$= \frac{2^{k+2} + 3(-1)^{k+1} + 2(-1)^k}{3 \cdot 2^{k+1}}$$

$$= \frac{-2^{k+2} + 3(-1)^{k+1} - (-2)(-1)^k}{3 \cdot 2^{k+1}}$$

$$= \frac{2^{k+2} + 3(-1)^{k+1} - (2)(-1)(-1)^k}{3 \cdot 2^{k+1}}$$

$$= \frac{2^{k+2} + 3(-1)^{k+1} - 2(-1)^{k+1}}{3 \cdot 2^{k+1}}$$

$$= \frac{2^{k+2} + 3(-1)^{k+1} - 2(-1)^{k+1}}{3 \cdot 2^{k+1}}$$

$$= \frac{2^{k+2} + (-1)^{k+1}}{3 \cdot 2^{k+1}}$$

True

Hence proved.

Question 1013

Prove that $1^2 - 2^2 + 3^2 - \dots + (-1)^{n-1} n^2 = \frac{(-1)^{n-1} n(n+1)}{2}$

Base Case :

$$1^2 - 2^2 + 3^2 - \dots + 3(-1)^{n-1} n^2 = \frac{(-1)^{n-1} n(n+1)}{2}$$

put $n = 1$

$$p(1) = 1^2 = \frac{(-1)^{1-1} n(n+1)}{2}$$

$$= 1^2 = \frac{(-1)^{1-1} 1(1+1)}{2}$$

$$= 1^2 = \frac{(-1)^0 1(2)}{2}$$

$$= 1^2 = \frac{(1)(2)}{2}$$

$$= 1^2 = 1$$

$1^2 = 1$ True.

Inductive step :

put $n = k$

$$1^2 - 2^2 + 3^2 - \dots + (-1)^{n-1} n^2 = \frac{(-1)^{n-1} n(n+1)}{2}$$

$$1^2 - 2^2 + 3^2 - \dots + (-1)^{k-1} (k^2) = \frac{(-1)^{k-1} k(k+1)}{2}$$

$$1^2 - 2^2 + 3^2 - \dots + (-1)^{k-1} k^2 = \frac{(-1)^{k-1} k(k+1)}{2}$$

$$1^2 - 2^2 + 3^2 - \dots + (-1)^{k-1} k^2 + (-1)^{k+1-1} (k+1)^2 = \frac{(-1)^{k+1-1} (k+1)(k+1)}{2}$$

$$1^2 - 2^2 + 3^2 - \dots + (-1)^{k-1} k^2 + (-1)^k (k+1)^2 = \left[\frac{(-1)^k (k+1)(k+2)}{2} \right]$$

$$\frac{(-1)^{k-1} k(k+1)}{2} + (-1)^k (k+1)^2$$

$$= \frac{(-1)^{k-1} k(k+1)}{2} + (-1)^k (k+1)^2$$

$$= \frac{(-1)^k (-1)^{-1} k(k+1)}{2} + (-1)^k (k+1)^2$$

$$= (-1)^k (k+1) \left[\frac{(-1)^{-1} (k)}{2} + (k+1) \right]$$

$$= (-1)^k (k+1) \left[\frac{(-1) (k)}{2} + (k+1) \right]$$

$$= (-1)^k (k+1) \left[\frac{-1(k)}{(1)(2)} + (k+1) \right]$$

$$= (-1)^k (k+1) \left[\frac{-k}{2} + (k+1) \right]$$

$$= (-1)^k (k+1) \left[\frac{-k + 2(k+1)}{2} \right]$$

$$= (-1)^k (k+1) \left[\frac{-k + 2k + 2}{2} \right]$$

$$= (-1)^k (k+1) \left[\frac{k+2}{2} \right]$$

$$= \left[\frac{(-1)^k (k+1)(k+2)}{2} \right] \text{ True}$$

Hence proved.

Question 2014 :-

Prove that for every positive Integer

$$n, \sum_{k=1}^n k 2^k = (n-1)2^{n+1} + 2$$

Soln:-

$$\sum_{k=1}^n k 2^k = (n-1)2^{n+1} + 2$$

$$(1)(2)^1 + 2(2)^2 + 3(2)^3 + 4(2)^4 + \dots + n \cdot 2^n = (n-1)2^{n+1} + 2$$

Base Case : $n=1$

$$P(1) = 1 \cdot 2^1 = (n-1)2^{n+1} + 2$$

$$1 \cdot 2^1 = (1-1)2^{1+1} + 2$$

$$2 = (0)2^2 + 2$$

$$2 = 0 + 2$$

$$2 = 2 \quad \text{True.}$$

Inductive Step :

$$1 \cdot 2^1 + 2 \cdot 2^2 + 3 \cdot 2^3 + \dots + n \cdot 2^n = (n-1)2^{n+1} + 2$$

$$n=k$$

$$1 \cdot 2^1 + 2 \cdot 2^2 + 3 \cdot 2^3 + \dots + k \cdot 2^k = (k-1)(2^{k+1}) + 2$$

$$1 \cdot 2^1 + 2 \cdot 2^2 + 3 \cdot 2^3 + \dots + k \cdot 2^k + (k+1)2^{k+1} = (k)(2^{k+2}) + 2$$

$$= (k-1)(2^{k+1}) + 2 + (k+1)(2^{k+1})$$

$$= [(k-1)(2^{k+1}) + (k+1)(2^{k+1})] + 2$$

$$= 2^{k+1} [(k-1) + (k+1)] + 2$$

$$= 2^{k+1} [k - 1 + k + 1] + 2$$

$$= 2^{k+1} [2k] + 2$$

$$\frac{(2^{k+1} \cdot 2)(k) + 2}{(2^{k+2})(k) + 2} \quad \text{True}$$

Hence proved.

Question 10 IS :-

Prove that for every positive Integer n
 $1 \cdot 2 + 2 \cdot 3 + \dots + n(n+1) = \frac{n(n+1)(n+2)}{3}$

Base Case : put $n=1$

$$P(1) = 1 \cdot 2 = \frac{n(n+1)(n+2)}{3}$$

$$= 1 \cdot 2 = \frac{1(1+1)(1+2)}{3}$$

$$= 1 \cdot 2 = \frac{1(2)(3)}{(3)}$$

$$2 = 2 \quad \text{True.}$$

Inductive Steps : $n=k$

$$1 \cdot 2 + 2 \cdot 3 + \dots + (n+1) = \frac{n(n+1)(n+2)}{3}$$

$$1 \cdot 2 + 2 \cdot 3 + \dots + k(k+1) = \frac{k(k+1)(k+2)}{3}$$

$$1 \cdot 2 + 2 \cdot 3 + \dots + (k)(k+1) + (k+1)(k+2) = \frac{(k+1)(k+2)(k+3)}{3}$$

$$\frac{k(k+1)(k+2)}{3} + (k+1)(k+2)$$

$$= \frac{k(k+1)(k+2) + 3(k+1)(k+2)}{3}$$

$$= \frac{(k+1)(k+2)(k+3)}{3} \quad \text{True}$$

Hence proved.

Question 16:

Prove that for every positive Integer n .

$$1 \cdot 2 \cdot 3 + 2 \cdot 3 \cdot 4 + \dots + n(n+1)(n+2) = \frac{n(n+1)(n+2)(n+3)}{4}$$

Base Case: put $n=1$

$$P(1) = 1 \cdot 2 \cdot 3 = \frac{n(n+1)(n+2)(n+3)}{4}$$

$$= 6 = \frac{(1)(1+1)(1+2)(1+3)}{4}$$

$$= 6 = \frac{(1)(2)(3)(4)}{4}$$

$$= 6 = \frac{24}{4}$$

$$6 = 6 \quad \text{True}$$

Inductive Step:

$$1 \cdot 2 \cdot 3 + 2 \cdot 3 \cdot 4 + \dots + n(n+1)(n+2) = \frac{n(n+1)(n+2)(n+3)}{4}$$

put $n=k$

$$1 \cdot 2 \cdot 3 + 2 \cdot 3 \cdot 4 + \dots + k(k+1)(k+2) = \frac{k(k+1)(k+2)(k+3)}{4}$$

$$1 \cdot 2 \cdot 3 + 2 \cdot 3 \cdot 4 + \dots + k(k+1)(k+2)(k+3) = \frac{(k+1)(k+2)(k+3)(k+4)}{4}$$

$$= \frac{k(k+1)(k+2)(k+3)}{4} + (k+1)(k+2)(k+3)$$

$$= \frac{k(k+1)(k+2)(k+3) + 4(k+1)(k+2)(k+3)}{4}$$

$$= \frac{(k+1)(k+2)(k+3)(k+4)}{4} \quad \text{True}$$

Hence proved.

Question 17 :-
Prove that $\sum_{j=1}^n j^4 = \frac{n(n+1)(2n+1)(3n^2+3n-1)}{30}$

whenever n is a positive integer.

$$(1)^2 + (2)^2 + (3)^2 + \dots + (n)^2 = \frac{n(n+1)(2n+1)(3n^2+3n-1)}{30}$$

Base Case : $n=1$

$$P(1) = (1)^4 = \frac{n(n+1)(2n+1)(3n^2+3n-1)}{30}$$

$$= (1)^4 = \frac{(1)(1+1)(2(1)+1)(3(1)^2+3(1)-1)}{30}$$

$$= 1 = \frac{(1)(2)(3)(5)}{30}$$

$$1 = \frac{30}{30}$$

$$1 = 1 \quad \text{True}$$

Inductive Steps :-

Re.H.S :-

$$\text{put } n=k$$

$$(1)^4 + (2)^4 + (3)^4 + \dots + (n)^4 = \frac{n(n+1)(2n+1)(3n^2+3n-1)}{30}$$

$$1^4 + 2^4 + 3^4 + \dots + k^4 = \frac{k(k+1)(2k+1)(3k^2+3k-1)}{30}$$

$$= \frac{(k+1)(k+2)(2k+3)(3k^2+3k-1)}{30}$$

$$= \frac{(k+1)(k+2)(2k+3)(3(k+1)^2+3k)}{30} \quad \text{--- (1)}$$

L.H.S :

$$\frac{k(k+1)(2k+1)(3k^2+3k-1)}{30} + (k+1)^4$$

$$\frac{k(k+1)(2k+1)(3k^2+3k-1) + 30(k+1)^4}{30}$$

$$= (k+1) \left[\frac{k(2k+1)(3k^2+3k-1)}{30} + 30(k+1)^3 \right]$$

$$\frac{k+1}{30} \left[(2k^2+k)(3k^2+3k-1) + 30(k^3+1+3k^2+3k) \right]$$

$$\frac{(k+1)[6k^4+6k^3-2k^2+3k^3+3k^2-k+30k^3+30+90k^2+90k]}{30}$$

$$= \frac{(k+1)[6k^4+39k^3+91k^2+89k+30]}{30} \quad \text{--- (ii)}$$

Further Solving eq (i)

$$\frac{(k+1)(k+2)(2k+3)(3(k+1)^2+3k+2)}{30}$$

$$= \frac{(k+1)(2k^2+3k+4k+6)(3(k^2+1+2k)+3k+2)}{30}$$

$$= \frac{(k+1)(2k^2+7k+6)(3k^2+3+6k+3k+2)}{30}$$

$$= \frac{(k+1)(6k^4+18k^3+10k^2+21k^3+63k^2+35k+18k^2+54k+30)}{30}$$

$$= \frac{(k+1)(6k^4+39k^3+9k^2+89k+30)}{30}$$

Now eq (i) and (ii)

$$R.H.S = L.H.S \text{ is True}$$

Hence proved.

Question 18:-

Let $P(n)$ be the statement that $n! < n^n$, where n is an integer greater than 1.

(a)

what is the statement $P(2)$?

n is an integer greater than 2.

let

$$P(n) = 2 \text{ Because } n \in 2^+ \{2, 3, 4, \dots\}$$

So,

$$P(2) = n! < n^n$$

$$P(2) = 2! < 2^2$$

$$= 2! < 4$$

$$= \boxed{2 < 4} \text{ True}$$

(b)

Show that $P(2)$ is true, Completing the basic step of the proof.

$$P(2) = 2! < 2^2$$

$$P(2) = 2! < 2^2$$

$$= \boxed{2 < 4} \quad \text{True}$$

(c)

What is the inductive hypothesis?

Inductive hypothesis:

put $n = k$

$$P(n) = n! < n^n$$

$$P(k) = k! < k^k$$

(d)

What do you need to prove in the inductive step?

Inductive step:

$$P(k) = k! < k^k$$

$$P(k+1) = (k+1)! < (k+1)^{k+1}$$

(e)

Complete the inductive step.

$$k! < k^k$$

To prove:-

$$(k+1)! < (k+1)^{k+1}$$

$$(k+1)! = (k+1) \times k!$$

So;

$$(k+1)! = (k+1) \times k! < (k+1)k^k$$

$$(k+1)! < (k+1)k^k$$

$$(k+1)! < (k+1)(k+1)^k \quad \because k < k+1$$

$$(k+1)! < (k+1)^{k+1}$$

$P(k+1)$ True

(f)

Explain why these steps show that this inequality is true whenever n is an integer greater than 1.

Both the basic step and Inductive steps are proved $\rightarrow$ True

So $(k+1)! < (k+1)^{k+1}$ proved

(b)

Question No. 19

Let $P(n)$ be the statement that

$$1 + \frac{1}{4} + \frac{1}{9} + \dots + \frac{1}{n^2} < 2 - \frac{1}{n}$$

where n is an integer greater than 1.

(a)

What is the statement $P(2)$?

$$P(n) = 1 + \frac{1}{4} + \dots + \frac{1}{n^2} < 2 - \frac{1}{n}$$

let $P(n) = 2$

So;

$$P(2) = 1 < \frac{2-1}{n}$$

$$= 1 < \frac{2-1}{2}$$

$$= 1 < \frac{4-1}{2}$$

$$= 1 < \frac{3}{2}$$

$$= \boxed{1 < 1.5} \text{ True}$$

(b)

Show that $P(2)$ is true, Completing the basic step of the proof.

$$P(2) = 1 < \frac{2-1}{2}$$

$$= 1 < \frac{4-1}{2}$$

$$= 1 < \frac{3}{2}$$

$$\boxed{1 < 1.5} \text{ True}$$

(c)

What is the Inductive hypothesis?

$$P(n) = 1 + \frac{1}{4} + \frac{1}{9} + \dots + \frac{1}{n^2} < \frac{2-1}{n}$$

$$P(k) = 1 + \frac{1}{4} + \frac{1}{9} + \dots + \frac{1}{k^2} < \frac{2-1}{k}$$

(d)

What do you need to prove in the inductive step?

$$P(k+1) = 1 + \frac{1}{4} + \frac{1}{9} + \dots + \frac{1}{k^2} + \frac{1}{(k+1)^2} < 2 - \frac{1}{k+1}$$

(e)

Complete the Inductive step.

$$P(k) = 1 + \frac{1}{4} + \frac{1}{9} + \dots + \frac{1}{k^2} < 2 - \frac{1}{k}$$

To prove

$$P(k+1) = 1 + \frac{1}{4} + \frac{1}{9} + \dots + \frac{1}{k^2} + \frac{1}{(k+1)^2} < 2 - \frac{1}{k+1}$$

$$1 + \frac{1}{4} + \frac{1}{9} + \dots + \frac{1}{k^2} + \frac{1}{(k+1)^2} < 2 - \frac{1}{k} + \frac{1}{(k+1)^2}$$

$$= 2 - \left[\frac{1}{k} - \frac{1}{(k+1)^2} \right] = 2 - \left[\frac{(k+1)^2 - k}{k(k+1)^2} \right]$$

$$= 2 - \left[\frac{k^2 + 2k + 1 - k}{k(k+1)^2} \right] = 2 - \left[\frac{k^2 + k + 1}{k(k+1)^2} \right]$$

$$= 2 - \left[\frac{k(k+1) + 1}{k(k+1)^2} \right] = 2 - \left[\frac{k(k+1) + 1}{k(k+1)^2} \right]$$

$$= 2 - \left[\frac{1}{(k+1)} + \frac{1}{k(k+1)^2} \right]$$

$$= 2 - \frac{1}{k+1} - \frac{1}{k(k+1)^2}$$

Neglecting $\frac{-1}{k(k+1)^2}$

So; It becomes,

$$= 2 - \frac{1}{k+1}$$

$$1 + \frac{1}{4} + \frac{1}{9} + \dots + \frac{1}{k^2} + \frac{1}{(k+1)^2} < 2 - \frac{1}{k+1} \quad \text{True.}$$

(If)
Explain why these steps show the inequality is true whenever n is an integer greater than 1.

Both the basic and Inductive steps are proved so the statement is True because

$$1 + \frac{1}{4} + \frac{1}{9} + \dots + \frac{1}{k^2} + \frac{1}{(k+1)^2} < 2 - \frac{1}{k+1} \quad \text{proved.}$$

Question No 20 :-

Prove that $3^n < n!$ if n is an integer greater than 6.

Base Case :

$$n \in \mathbb{Z}^+ = \{7, 8, 9, \dots\}$$

$$\text{let } P(n) = 3^n < n!$$

$$P(7) = 3^7 < 7!$$

$$P(7) = 3^7 < 7!$$

$$= 2187 < 5040 \quad \text{True.}$$

Inductive Step :

put $n=k$

$$P(k) = 3^k < k!$$

$$P(k+1) = 3^{k+1} < (k+1)!$$

$$3^k < k!$$

Multiplying by $(k+1)$ on Both sides:

$$(k+1)3^k < (k+1) \cdot k!$$

$$(k+1)3^k < (k+1)!$$

$$(k-2+3)3^k < (k+1)!$$

$$[k-2+(3)]3^k < (k+1)!$$

$$(k-2)3^k + 3 \cdot 3^k < (k+1)!$$

$$(k-2)3^k + 3^{k+1} < (k+1)!$$

Negat

$$(k-2)3^k - 8$$

$$3^{k+1} < (k+1)!$$

$P(k+1)$ is True

Hence proved.

Question 10 Q1 :-

Prove that $2^n > n^2$ if n is an integer greater than 4.

Base Case :

$$\text{Let } P(n) = 5$$

$$n \in 2^+ \{5, 6, 7, \dots\}$$

$$P(5) = 2^5 > 5^2$$

$$= 2^5 > 5^2$$

$$32 > 25$$

True

Inductive Case :

$$\begin{aligned} \text{put } n &= k \\ P(k) &= 2^k > k^2 \\ &= 2^k > k^2 \end{aligned}$$

$$P(k+1) = 2^{k+1} > (k+1)^2$$

As

$$P(k) = 2^k > k^2$$

To prove that

$$2^{k+1} > (k+1)^2$$

$$(k+1)^2 = k^2 + 2k + 1$$

$\because k \geq 1$

$$(k+1)^2 < k^2 + 2k + k$$

$$(k+1)^2 < k^2 + 3k$$

$\because 3 < k$

$$(k+1)^2 < k^2 + k \cdot k$$

$\because 2^k > k^2$

$$(k+1)^2 < 2k^2$$

$$(k+1)^2 < 2^{k+1}$$

$$2^{k+1} > (k+1)^2$$

$P(k+1)$ is true

Both Base and Inductive steps are true so By mathematical Induction $P(k+1) = 2^{k+1} > (k+1)^2$

True. Hence proved

Question No 22 :-

For which nonnegative integers n is $n^2 \leq n!$?
Prove your answer.

$$n \in \mathbb{Z}^+ = \{0, 1, 2, 3, \dots\}$$

$$n=0 \Rightarrow P(0) = 0^2 \leq 0! \Rightarrow 0 \leq 1 \quad (\text{True})$$

$$n=1 \Rightarrow P(1) = 1^2 \leq 1! \Rightarrow 1 \leq 1 \quad (\text{True})$$

$$n=2 \Rightarrow P(2) = 2^2 \leq 2! \Rightarrow 4 \not\leq 2 \quad (\text{but } 4 > 2)$$

$$n=3 \Rightarrow P(3) = 3^2 \leq 3! \Rightarrow 9 \not\leq 6 \quad (\text{but } 9 > 6)$$

$$n=4 \Rightarrow P(4) = 4^2 \leq 4! \Rightarrow 16 \leq 24 \quad (\text{True})$$

$$n=5 \Rightarrow P(5); 5^2 \leq 5! \Rightarrow 25 \leq 120 \text{ True}$$

So;

$$P(n) = n^2 \leq n!; n \geq 4$$

Base Case :

$$\text{Put } n=4$$

$$P(n) = n^2 \leq n!$$

$$P(4) = 4^2 \leq 4!$$

$$P(4) = 16 \leq 24 \text{ True.}$$

Inductive Step :

$$\text{Put } n=k$$

$$P(k) = k^2 \leq k!$$

$$P(k+1) = (k+1)^2 \leq (k+1)!$$

As :

$$k^2 \leq k!$$

Prove :-

$$(k+1)^2 \leq (k+1)!$$

$$k^2 \leq k!$$

Multiply $k+1$ on Both sides

$$(k+1)k^2 \leq (k+1)k!$$

$$k^3 + k^2 \leq (k+1)!$$

$$k^2 + 2k - 2k + 1 - 1 + k^3 \leq (k+1)!$$

$$(k^2 + 2k + 1) + (k^3 - 2k - 1) \leq (k+1)!$$

$$(k+1)^2 \leq (k+1)!$$

$\because k^3 - 2k - 1$ is +ve
ignore it.

$P(k+1)$ is true

Hence proved.

Question 10 23 :-

For which non-negative integers n is $2n+3 \leq 2^n$? prove your answer.

Sol :-

$$\text{Let } P(n) = 2n+3 \leq 2^n$$

$$n \in \mathbb{N} \setminus \{0, 1, 2, 3, \dots\}$$

$$n=0 \Rightarrow P(0) = 2(0)+3 \leq 2^0 \Rightarrow 3 \leq 1 \quad (F)$$

$$n=1 \Rightarrow P(1) = 2(1)+3 \leq 2^1 \Rightarrow 5 \leq 2 \quad (F)$$

$$n=2 \Rightarrow P(2) = 2(2)+3 \leq 2^2 \Rightarrow 7 \leq 4 \quad (F)$$

$$n=4 \Rightarrow P(4) = 2(4)+3 \leq 2^4 \Rightarrow 11 \leq 16 \quad (F)$$

$$n=5 \Rightarrow P(5) = 2(5)+3 \leq 2^5 \Rightarrow 13 \leq 32 \quad (T)$$

$$n=3 \Rightarrow P(3) = 2(3)+3 \leq 2^3 \Rightarrow 9 \leq 8 \quad (F)$$

$$P(n) = 2n+3 \leq 2^n$$

Basic Step :

put $n=4$

$$P(4) = 2(4)+3 \leq 2^4 \Rightarrow 11 \leq 16 \quad \text{True}$$

Inductive Step :

put $n=k$

$$P(k) = 2k+3 \leq 2^k$$

$$P(k+1) = (2k+1)+3 \leq 2^{k+1}$$

To prove

$$2(k+1)+3 \leq 2^{k+1}$$

$$2(k+1)+3 = 2k+2+3$$

$$= (2k+3) + 2$$

$$\leq 2^k + 2$$

$$< 2^k + 2^k$$

$$< 2^k + 2^k$$

$$< 2^{k+1}$$

$$2(k+1)+3 < 2^{k+1}$$

$P(k+1)$ is True

Hence proved

Question 20 (248-)

Prove that $1/(2n) \leq [1 \cdot 3 \cdot 5 \cdots (2n-1)] / (2 \cdot 4 \cdots 2n)$ whenever n is a positive integer.

$$n \in \mathbb{Z}^+ = \{1, 2, 3, \dots\}$$

Base Case:

$$P(1) \equiv \frac{1}{2(1)} \leq \frac{1}{2} = \frac{1}{2} \quad \text{True}$$

Inductive Step:

Put $n = k$

Then for $2k-1+2$

2^{k+1}

$$\text{Let } P(k) = \frac{1}{2^k} \leq \frac{1 \cdot 3 \cdot 5 \cdots (2k-1)}{2 \cdot 4 \cdot 6 \cdots (2k)} \quad (1)$$

$$\text{To prove } P(k+1) = \frac{1}{2(k+1)} \leq \frac{1 \cdot 3 \cdot 5 \cdots (2k-1)(2k+1)}{2 \cdot 4 \cdot 6 \cdots (2k)(2k+2)}$$

$$\frac{1 \cdot 3 \cdot 5 \cdots (2k-1)(2k+1)}{2 \cdot 4 \cdot 6 \cdots (2k)(2k+2)} \geq \frac{1}{2(k+1)}$$

$$\geq \left(\frac{2k+1}{2k} \right) \left(\frac{1}{2(k+1)} \right)$$

$$\geq \left(\frac{2k+1}{2k} \right) \left(\frac{1}{2(k+1)} \right)$$

$$\geq \left(1 + \frac{1}{2k}\right) \cdot \left(\frac{1}{2(k+1)}\right)$$

As

$$1 + \frac{1}{2k} > 1$$

$$\geq (1) \cdot \left(\frac{1}{2(k+1)}\right)$$

So;

$$\frac{1 \cdot 3 \cdot 5 \cdots (2k-1)(2k+1)}{2 \cdot 4 \cdot 6 \cdots (2k)(2k+2)} \geq \frac{1}{2(k+1)}$$

$P(k+1)$ is true

Hence proved.

Question No 31:-

Prove that 2 divides $n^2 + n$ whenever n is a positive integer.

[Base Case :-] put $n = 1$

$$p(n) = n^2 + n$$

$$p(1) = (1)^2 + 1$$

$$p(1) = 1 + 1$$

$$= 2 \quad \text{True}$$

[Inductive Step :-]

put $n = k$

$$p(n) = n^2 + n$$

$$p(k) = k^2 + k$$

$$p(k+1) = (k+1)^2 + (k+1)$$

To prove:

$(k+1)^2 + k+1$ is divisible by 2

So.

$$(k+1)^2 + k+1 = k^2 + 2k + 1 + k+1$$

$$= k^2 + k + 2k + 2$$

$$= (k^2 + k) + 2(k+1)$$

As $k^2 + k$ is divisible by 2

So,

$$(k^2 + k) + 2(k+1) = 2m + 2(k+1)$$

$$= 2(m + k + 1)$$

Since it's clearly divisible by 2 so the
 $P(k+1)$ is true by mathematical Induction.

(Question ~ 32 a-

Prove that 3 divides $n^3 + 2n$ whenever n is a positive Integer.

Base Case :-

let $n = 1$

$$p(n) = 1^3 + 2(1)$$

$$= 1 + 2$$

$$= 3 \quad \text{True}$$

Inductive Step :-

put $n = k$

$$p(k) = k^3 + 2(k)$$

$$p(k+1) = (k+1)^3 + 2(k+1)$$

To prove that :

$(k+1)^3 + 2(k+1)$ is divisible by 3.

So,

$$(k+1)^3 + 2(k+1) = (k^3 + 1 + 3k + 3k^2) + 2k + 2$$

$$= k^3 + 3k^2 + 3k + 1 + 2k + 2$$

$$= k^3 + 3k^2 + 3k + 2k + 3$$

$$= k^3 + 3k^2 + 5k + 3$$

$$= k^3 + 3k^2 + 3k + 2k + 3$$

$$= k^3 + 2k + 3k^2 + 3k + 3$$

As $k^3 + 2k$ is divisible by 3 So

$$k^3 + 2k = 3(m)$$

Now e-

$$= 3m + 3k^2 + 3k + 3$$

$$= 3m + 3(k^2 + k + 1)$$

Both the sides are clearly divisible by 3 by mathematical Induction its True.

Hence proved.

Question 10 33 e-

Prove that 5 divides $n^5 - n$ wherever n is a non-negative Integer.

Base Case e-

put $n = 0$

$$p(n) = n^5 - n$$

$$p(0) = (0)^5 - 0$$

$$p(0) = 0 - 0$$

$$= 0 \quad \text{True}$$

5 is divisible by 0.

Inductive Step e

put $n = k$

$$p(k) = k^5 - k$$

$$p(k+1) = (k+1)^5 - (k+1)$$

To prove that

$$(k+1)^5 - (k+1)$$

General Algebraic Formula :-

$$(a+b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$$

So;

$$= (k+1)^5$$

$$= (a+b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$$

$$\because \binom{n}{k} = \frac{n!}{k!(n-k)!}$$

$$= \binom{5}{0} k^{5-0} (1)^0 + \binom{5}{1} k^{5-1} (1)^1 + \binom{5}{2} k^{5-2} (1)^2 \\ + \binom{5}{3} k^{5-3} (1)^3 + \binom{5}{4} k^{5-4} (1)^4 + \binom{5}{5} k^{5-5} (1)^5$$

Now $(k+1)^5 = (k+1) \cdot$ because

$$(1) k^5 + 5k^4 + 10k^3 + 10k^2 + 5k + 1(k+1)$$

$$= k^5 + 5k^4 + 10k^3 + 10k^2 + 5k + 1 - k - 1$$

$$= k^5 + 5k^4 + 10k^3 + 10k^2 + 5k - k$$

$$= k^5 - k + 5k^4 + 10k^3 + 10k^2 + 5k$$

$$= (k^5 - k) + 5(k^4 + 2k^3 + 2k^2 + k)$$

As

$k^5 - k$ is divisible by 5.

$$5(m) + 5(k^4 + 2k^3 + 2k^2 + k) \text{ True}$$

Divisible by 5 which is clearly visible

Hence proved by mathematical Induction.

Question No 34 :-

Prove that 6 divides $n^3 - n$ whenever

n is a non-negative integers.

Base Case :- put $n = 0$

$$p(n) = n^3 - n$$

$$p(0) = (0)^3 - 0$$

$$p(0) = 0 \text{ divisible by 6 True.}$$

Inductive step :-

put $n = k$

$$p(k) = k^3 - k$$

$$p(k+1) = (k+1)^3 - (k+1)$$

To prove :-

So: $(k+1)^3 - (k+1)$ Divisible by 5

$$\begin{aligned} & (k+1)^3 - k - 1 \\ &= (k^3 + 1 + 3k^2 + 3k) - (k+1) \\ &= k^3 + 3k^2 + 3k + 1 - k - 1 \\ &= k^3 + 3k^2 + 3k - k \\ &= (k^3 - k) + 3k^2 + 3k \\ &= (k^3 - k) + 3(k^2 + k) \end{aligned}$$

$$\begin{aligned} \text{As } & (k^3 - k) + 3(k^2 + k) \\ &= 6m + 3(k^2 + k) \\ &= 3(2m + k^2 + k) \quad \text{True} \end{aligned}$$

By Mathematical Induction 6 is divisible with $3(2m + k^2 + k)$ so, True. Hence proved.

* Question 35 :-

Prove that $n^2 - 1$ is divisible by 8 whenever n is an odd positive integer.

Soln:-

Base Case:

put $n = 1$

$$p(n) = n^2 - 1$$

$$p(n) = (2n-1)^2 - 1$$

$$p(1) = (2(1)-1)^2 - 1$$

$$p(1) = (1)^2 - 1$$

$$p(1) = (0) \quad \text{True}$$

Divisible by 8

Inductive Step:

put $n = k$

$$p(k) = (2k-1)^2 - 1$$

For next odd integers

$$p(k+1) = (2(k+1)-1)^2 - 1$$

$$= (2k+2-1)^2 - 1$$

$$= (2k+1)^2 - 1$$

Simplify these two expressions by difference
Comparising:

$$\begin{aligned}
 & ((2k+1)^2 - 1) - ((2k-1)^2 - 1) \\
 &= (2k+1)^2 - 1 - (2k-1)^2 + 1 \\
 &= \frac{(2k+1)^2}{b} - \frac{(2k-1)^2}{a} \quad \because a^2 - b^2 \\
 &= (2k+1) + (2k-1)(2k+1) - (2k-1) \\
 &= (2k+1+2k-1)(2k+1-2k+1) \\
 &= (4k)(2) \\
 &= \boxed{8k}
 \end{aligned}$$

True

Also:

$(2k-1)^2 - 1$ is also divisible by 8
So the sum of the $8k$ and $(2k-1)^2 - 1$ is
also divisible by 8 True

Hence proved.

* Question 36 :-

Prove that 21 divides $4^{n+1} + 5^{2n-1}$ whenever
 n is a positive integer.

Soln:-

Base Case :

$$\begin{aligned}
 & \text{put } n = 1 \\
 & p(n) = 4^{n+1} + 5^{2n-1} \\
 & p(1) = 4^{1+1} + 5^{2(1)-1} \\
 & p(1) = 4^2 + 5^{2-1}
 \end{aligned}$$

$$16 + 5$$

$$= 21$$

True divisible by 21.

Inductive Steps :

put $n = k$

$$p(k) = 4^{k+1} + 5^{2k-1}$$

$$p(k) = 4^{k+1} + 5^{2k-1}$$

$$p(k+1) = 4^{k+2} + 5^{2(k+1)-1}$$

$$p(k+1) = 4^{k+2} + 5^{2k+2-2}$$

$$4^{k+2} + 5^{2k+1}$$

To prove that

$4^{k+2} + 5^{2k+1}$ is divisible by 21

$$(4^{k+2} + 5^{2k+1}) - (4^{k+1} + 5^{2k-1})$$

$$= 4^{k+2} - 4^{k+1} + 5^{2k+1} - 5^{2k-1}$$

$$= (4^{k+2} - 4^{k+1}) + (5^{2k+1} - 5^{2k-1})$$

Simplify :

$$4^{k+2} - 4^{k+1}$$

$$= 4^{k+1} \cdot 4 - 4^{k+1} \cdot 1$$

$$= 4^{k+1} (4 - 1)$$

$$= 4^{k+1} (3)$$

Simplifying :

$$5^{2k+1} - 5^{2k-1}$$

$$= 5^{2k-1} \cdot 5^2 - 5^{2k-1}$$

$$= 5^{2k-1} (5^2 - 1)$$

$$= 5^{2k-1} (24)$$

The term becomes

$$4^{k+1} (3) + (5^{2k-1} (24))$$

$$= 3(4^{k+1} + 8 \cdot 5^{2k-1})$$

The expression is divisible by 3

when $k=1$

$$4^{k+2} + 8(5^{2k-1})$$

$$= 4^{1+1} + 8(5^{2 \cdot 1 - 1})$$

$$= 4^2 + 8(5^1)$$

$$16 + 8(5)$$

$$= 56$$

$$56 \div 7 = 8 \text{ Divisible by 7}$$

When $k=2$

$$= 4^{k+1} + 8(5^{2k-1})$$

$$= 1054 = 1064 \div 7 = 152$$

Divisible by 7

when $k=3$

$$= 4^{k+1} + 8(5^{2k-1})$$

$$= 25256$$

$$= 25256 \div 7 = 3614 \text{ Divisible by 7}$$

Since the expression is divisible by $[3]$ and $[7]$ so it is also means divisible by

$[21]$ True

Hence proved.

* Question 37 :-

Prove that if n is a positive Integer then 133 divides $11^{n+1} + 12^{2n-1}$

Base Case :

put $n=1$

$$p(n) = 11^{n+1} + 12^{2n-1}$$

$$= 11^{1+1} + 12^{2(1)-1}$$

$$= 11^2 + 12^1$$

$$= 121 + 12$$

$[133]$ True

Divisible by 133.

Inductive Steps :

put $n=k$

$$p(k) = 11^{k+1} + 12^{2k-1}$$

$$p(k+1) = 11^{k+2} + 12^{2(k+1)-1}$$

$$= 11^{k+2} + 12^{2k+2-1}$$

$$= 11^{k+2} + 12^{2k+1}$$

To prove that

So, $11^{k+2} + 12^{2k+1}$ divisible by 133

$$\begin{aligned} & 11^{k+2} + 12^{2k+1} \\ &= 11 \cdot 11^{k+1} + 12^2 \cdot 12^{2k-1} \\ &= 11 \cdot 11^{k+1} + (11 + 133) \cdot 12^{2k-1} \\ &= 11 \cdot 11^{k+1} + (12^{2k-1} \cdot 11) + (133 \cdot 12^{2k-1}) \\ &= \boxed{11(11^{k+1} + 12^{2k-1}) + 133 \cdot 12^{2k-1}} \end{aligned}$$

The whole expression is divisible by 133 $\rightarrow$ True
Hence proved.